

Abderraouf Maoudj, Ph.D.

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About

📖 Dr. **Abderraouf Maoudj** holds a Ph.D. in Electronics & Robotics and specializes in multi-robot coordination, robotic manipulation and learning, and AI-driven control. Across academia and industry, he has led multidisciplinary projects that deliver field-ready robotic solutions, bridging high-impact research with hands-on prototyping and system integration. He has a proven record of project leadership and collaboration with companies such as Siemens, Sonacome (SNVI), and Enable Robotics, translating advanced robotics and AI technologies into practical automation systems. Combining deep technical expertise with a clear understanding of industrial needs, he delivers innovations that enhance efficiency, safety, and performance in manufacturing and beyond.

Currently, Dr. Maoudj is a *Postdoctoral Research Fellow* at the Interdisciplinary Research Center for Intelligent Manufacturing & Robotics (IRC-IMR), King Fahd University of Petroleum and Minerals (KFUPM), where he leads a programme advancing robotics through continuous learning and adaptation. Previously, he spent three years as a Postdoctoral Researcher at the University of Southern Denmark (SDU) and seven years as a Senior Researcher at the Center for the Development of Advanced Technologies (CDTA) in Algeria, consistently bridging the gap between academic research and industrial deployment.

His research focuses on enabling large teams of autonomous robots to coordinate and complement each other through real-time, localized decision-making. By leveraging decentralized control and continual learning, his work empowers individual robots to act independently while integrating seamlessly into larger systems—boosting efficiency and scalability in logistics, automated warehousing, manufacturing, and transportation.

Since 2014, he has also taught and mentored students in robotics and AI, designing curricula aligned with industry requirements and providing hands-on, practical education that prepares graduates to excel in the rapidly evolving robotics sector.

Education

- 2013 – 2018 📖 **Ph.D., University of Sciences and Technology Houari Boumediene (USTHB), Electronics & Robotics.**
Thesis title: *Remote Control of a Heterogeneous Multi-Robot System: Application to the RobuTER/ULM and AGVM6 Robots.*
- 2010 – 2012 📖 **Master's Degree, Mohamed Seddik Ben Yahia University of Jijel, Electronics.**
Thesis title: *Design and Realization of a Mobile Robot.*
- 2007 – 2010 📖 **B.Sc., Mohamed Seddik Ben Yahia University of Jijel, Electrical Engineering.**
Thesis title: *Realization of a Card Reader.*

Work Experiences

- Oct. 2023 – Present  **Postdoctoral Fellow, Interdisciplinary Research Center for Intelligent Manufacturing & Robotics (IRC-IMR), King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia.** (i) Leading a research project on AI-driven continuous learning and adaptation in collaborative robotics and human-robot interaction. (ii) Developing cognitive capabilities for multi-arm robotic systems to enhance flexibility and autonomy. (iii) Built Gazebo/PyBullet simulators and DRL pipelines (SAC, MAPPO).
- Dec. 2020 – Oct. 2023  **Postdoctoral Fellow, SDU Biorobotics, Mads Clausen Institute, University of Southern Denmark (SDU), Denmark.** (i) Conducted research on swarm robotics and multi-robot coordination methods for Industry 4.0 applications, focusing on warehouse environments.
- Oct. 2014 – Dec. 2020  **Senior Researcher, Robotics Division, Center for the Development of Advanced Technologies (CDTA), Algeria.** (i) Developed advanced robotic solutions, bridging academic research with practical industrial applications. (ii) Directed a cyber-physical Industry 4.0 platform combining PLC assembly cells, RFID part tracking, and AGV fleets. (iii) Designed PIC/STM32 motion controllers and real-time PID loops; built Raspberry Pi ↔ Arduino comms for aquatic and ground robots. (iv) Pioneered AI-based navigation (Q-learning, GA-optimised MPC), improving inspection-robot coverage. (v) Managed budgets, secured grants, and forged collaborations with Siemens and Sonacome (SNVI). (vi) Mentored junior engineers; delivered in-house training on sensor conditioning, signal processing, and ROS.

Research and Development Projects

- Current Project  **Advancing Robotics for Industry 4.0: Integrating Collaborative Robots to Enhance Industrial Tasks (IMR Lab, KFUPM):** The primary goal of my project is to develop AI-driven solutions that address key industrial needs and streamline the integration of collaborative robotic systems into smart manufacturing environments. While the ongoing waves of advancement in AI have significantly improved robot autonomy and manipulation, sophisticated human-robot collaboration and dynamic adaptation remain critical challenges. To bridge this gap, we equip manipulator robots with cognitive capabilities—leveraging continuous learning, visual learning methods, and transformer-based architectures for multi-arm coordination—so they can execute complex, varied tasks with human-like adaptability and problem-solving skills. By enhancing efficiency, flexibility, and service quality, this project aspires to drive transformative advancements in robotics and automation across modern production lines.

Research and Development Projects (continued)

Previous Projects

- **Swarm Robotics in Industry 4.0 (Bio-Robotics Lab):** Centering on the decentralized control of intelligent robot fleets pertinent to Industry 4.0, this research developed efficient coordination methods for large fleets in industrial settings. As a post-doc, my research in this project focused on the decentralized coordination of numerous robots, at the interface between conventional control and artificial intelligence. We bridged the gap between simulation and real-world application, enhancing robot negotiation and conflict resolution in dynamic environments. As an outcome of this project, we contributed to the state of the art by providing a fast, scalable, and robust decentralized planner that can coordinate up to a thousand mobile robots in large warehouses. A version of the software is available in GitHub Repository: <https://github.com/Amaoudj/ANTS2022.git> (Video Resource: <https://youtu.be/PrhIZ_mQp5I>).

A distinctive feature of this project was the introduction of a capacitated multi-robot scheduling system, aiming to boost efficiency in logistics, warehouses, and transportation sectors. This system was designed to optimize the allocation of product transport tasks and the planning of routes for AGVs, taking into account both capacity limitations and compatibility requirements.

- **Industry 4.0 Platform Development (Robotic and Manufacturing, CDTA):** This project entailed designing and implementing advanced algorithms and control systems for Industry 4.0 platforms, integrating robotic concepts and smart automation. Key tasks included: - Designing collaborative tasks with Java and KUKA LBR iiwa 7R800 integration. - Developing a RFID-based tracking system for assembly parts, integrated with PLC systems. - Creating a microcontroller-based motion control system for AGVs and a software application for robot task control and monitoring (C#, Arduino, Hokoyo). - Developing a navigation system with obstacle avoidance using reinforcement learning.
- **Remote Control of Heterogeneous Robots Advisor (Robotic and Manufacturing, CDTA):** This project involved: - Developing a distributed multi-agent system for scheduling and control in industrial robot fleets. - Creating path optimization methods using AI algorithms like Reinforcement-learning. - Designing a Fuzzy-based controller for mobile manipulator robots. - Developing collaboration mapping and a fault-tolerant multi-agent scheme for leader/follower control in homogeneous mobile robot fleets.
- **Aquatic Boat Robot for Bathymetry (Robotic and Manufacturing, CDTA):** This project aimed to provide a secure solution for bathymetric measurement using a remotely controlled boat robot. My role focused on the electronic realization of the robot.

Complete list of Publications

<https://scholar.google.com/citations?user=Fn2egmsAAAAJ>



Skills

Robotics	🔖 Robotic Programming: Experienced with a variety of robotic platforms, including MiR mobile robots, Enable mobile manipulator robots, Mecaademic 500 robotic arms (V3 and V4), and KUKA LBR iiwa 7R800 arms. Adept at programming real robots using Python and C/C++, complemented by ROS programming in Python.
Electronics	🔖 Electronic Design: Skilled in simulation and creation of control systems utilizing PIC microcontrollers, Arduino, and Raspberry Pi. Proficient in sensor conditioning, industrial electronics, and PLC programming with Siemens Tia-Portal.
Languages	🔖 Strong reading, writing, and speaking competencies for English, Arabic, and French.
Coding	🔖 Programming: Python, R, C, C++, Java, C#, HTML
Miscellaneous	🔖 Academic: Engaged in academic research and teaching, with a focus on applied sciences, Robotics and technology.
Areas of Research Interest	🔖 Robotics, Artificial Intelligence, Smart Automation and Optimization, Multi-agent Deep Reinforcement Learning, Multi-Robot Systems, Decentralized Multi-Agent Control, and Industry 4.0.

References

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